Bouncing Ball Game

Clone Repo: https://github.com/codertheashish/Bouncing-Ball

1. Introduction

The **Bouncing Ball Game** is a simple arcade game developed using **Python** and the **Pygame** library. In this game, the player controls a paddle at the bottom of the screen and keeps the ball bouncing. Every successful bounce increases the score. After every five points, the game level increases and the ball moves faster, making the game more challenging. If the player misses the ball, the game resets and starts again.

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2. Purpose

The main purpose of this project is to learn the basics of **2D game development** and understand object movement and collision handling using Pygame.

This project helps in learning:

- Game loop
- Paddle movement
- Ball physics
- Collision detection
- Score system
- Level progression
- Speed control
- Keyboard input handling

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3. Technologies Used

Technology	Purpose
Python	Main programming language
Pygame	Creates the game window, graphics, and keyboard controls
Random Module	Generates the initial direction of the ball

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4. Main Features

✓ Speed Selection Menu

Before the game starts, the player can choose one of three speed modes:

- Slow
- Normal
- Fast

The selected speed decides the starting speed of the ball, allowing the player to choose the difficulty level.

✓ Paddle Movement

The player controls the paddle using the keyboard.

Controls:

- **Left Arrow** → Move paddle left
- **Right Arrow** → Move paddle right

The paddle cannot move outside the game window, ensuring smooth gameplay.

✓ Ball Movement and Physics

The ball moves automatically after the game starts.

- The ball bounces from the left and right walls.
- The ball also bounces from the top wall.
- When the ball touches the paddle, it changes direction and continues moving.

This creates continuous and smooth gameplay.

✓ Score System

Every time the player successfully hits the ball with the paddle:

- The score increases by **1**.
- The updated score is displayed on the screen.

This allows the player to track their performance during the game.

✓ Level System

The game becomes more difficult as the player scores more points.

- Every **5 points**, the level increases.
- The speed of the ball also increases.

This gradually increases the challenge and keeps the game interesting.

✓ Game Reset

If the player misses the ball and it falls below the screen:

- The ball returns to the center.
- The score resets to **0**.
- The level resets to **1**.
- The game starts again with the selected base speed.

This allows the player to restart immediately without reopening the game.

✔ Exit Option

The player can press the **ESC** key at any time to close the game safely.

Summary

"This project is a Bouncing Ball game developed using Python and the Pygame library. The player controls a paddle to keep the ball bouncing and earn points. The game includes a speed selection menu, score system, level progression, increasing ball speed, collision detection, and automatic game reset when the ball is missed. Through this project, I learned game loops, keyboard event handling, collision detection, and object movement using Pygame."